

Designing a Galvanically Isolated Gate Driver

An optocoupled MOSFET driver with split control/power grounds, built for a Formula SAE power stage — five design revisions, one datasheet-only process.

1. Context and Goal

High-side and low-side power switching on the car (motor controllers, contactors, pump/fan drivers) is exactly where a noisy, high-current power stage can feed back into the low-voltage control electronics if the two are not properly separated. This module drives an IRF3205 power MOSFET from a 3.3 V/5 V PWM signal while keeping the MCU's control electronics galvanically isolated from the 12 V switching side, so a fault or transient on the power path cannot ride back into the MCU rail.

As with the rest of the portfolio, the design was built directly from manufacturer datasheets — Toshiba's TLP250 optocoupler gate-driver datasheet and Infineon's IRF3205 MOSFET datasheet — with no third-party reference boards. Every resistor value, the diode choice, and the ground-splitting strategy trace back to a specific recommendation in one of those two documents.

2. Final Design

A TLP250 optocoupler gate driver sits at the center of the board and is the actual isolation barrier: its LED input is driven by the PWM signal through R3 (330 Ω) on the control side, and its output stage drives the IRF3205 gate on the power side through R2 (10 Ω) gate resistor, with R1 (10 k Ω) as a gate pull-down. D1 is the flyback diode across the switched inductive load, and C1 (100 nF) decouples the TLP250's +12 V supply pin right at the package.

The detail that matters most for a power-stage board like this: **the two sides of the optocoupler are given two physically separate ground pools. GND** is the control/MCU-referenced ground, carrying only the PWM input and the LED-side return of the TLP250. **GNDP** is the power ground, carrying the +12 V gate-drive rail, the IRF3205 source path, and the flyback diode return. The two pools never share copper anywhere on the board — the only coupling between them is the optical link inside the TLP250 itself. This keeps switching noise, ground bounce, and any fault current on the 12 V power path from ever appearing on the MCU's reference, which is the entire point of using an optocoupled driver instead of a direct gate-drive transistor.

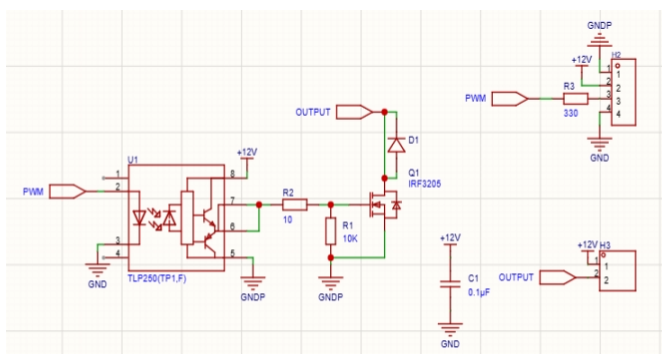


Fig. 1 — Final schematic: TLP250 isolation barrier, split GND / GNDP references either side.

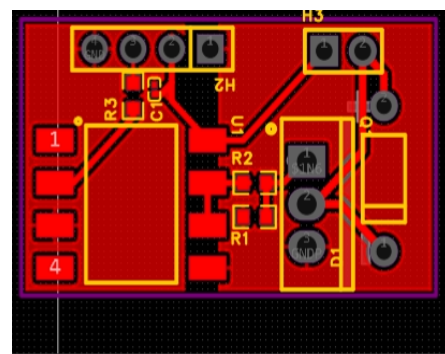


Fig. 2 — Final routed PCB layout, GND and GNDP kept as separate copper regions.

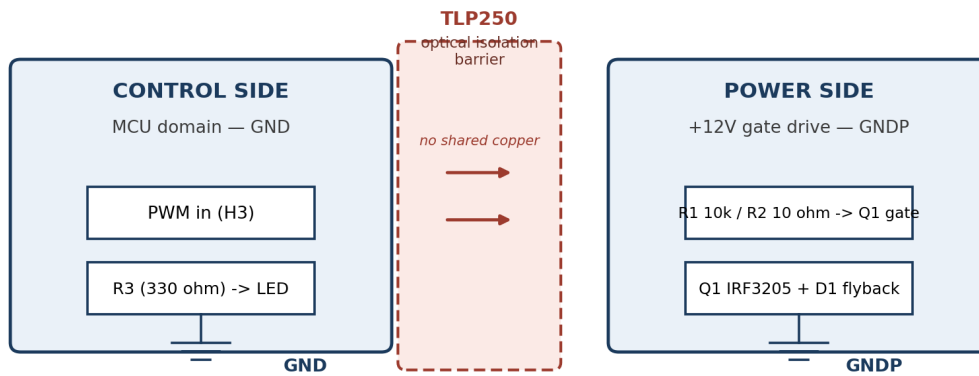


Fig. 3 — Ground-splitting strategy: GND (control) and GNDP (power) are only linked through the TLP250's optical barrier, with no shared copper.

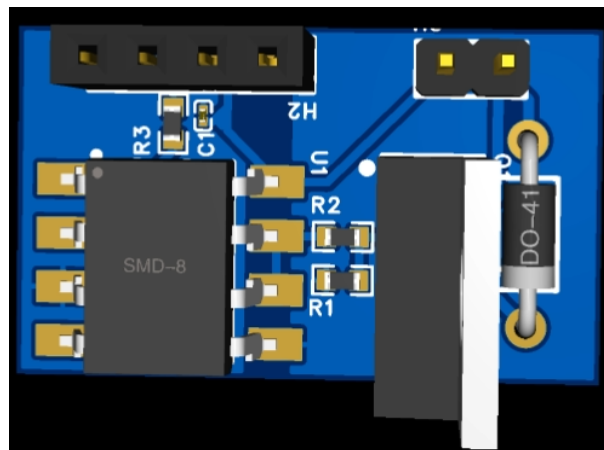


Fig. 4 — 3D render of the finished board: TLP250 optocoupler (SMD-8, left) driving the IRF3205 MOSFET (TO-220, right) with the flyback diode alongside.

3. Five Revisions

The board that made it into the portfolio is revision 5. Two schematic-level mistakes and three layout mistakes were caught and fixed along the way — kept here because they're more useful than the final schematic on its own.

Schematic revisions

Rev. 1 — No gate pull-down resistor

The very first schematic connected the TLP250 output pin straight to the IRF3205 gate with only the series gate resistor, no pull-down to GNDP. With nothing holding the gate low while the optocoupler's output stage is in a high-impedance state (at power-up, or if the LED-side drive is momentarily lost), the gate can float and let the MOSFET partially turn on by itself. R1 (10 kΩ) was added in revision 2 to guarantee the gate is pulled to GNDP whenever the driver isn't actively asserting it — a fix taken directly from the TLP250 application notes on driving a MOSFET gate.

Rev. 2 — No flyback path for the switched load

The second schematic pass still assumed a purely resistive load and left out any protection for the inductive kickback a motor or relay coil produces when the MOSFET turns off. Without D1, that turn-off transient shows up as a voltage spike across the drain-source junction, well past the IRF3205's rated VDS in a worst case. D1 was added across the load as a freewheeling diode, sized from the IRF3205 datasheet's safe-operating-area guidance for the expected switching current.

PCB revisions

Rev. 3 — GND and GNDP planes bridged by a stray trace

The first layout pass routed a decoupling capacitor's return path across both ground pools to save a routing detour, effectively short-circuiting the isolation the TLP250 was there to provide. It passed a visual schematic check but defeated the entire point of the split-ground design. Caught by tracing every GND and GNDP net individually before moving to revision 4 — no trace, via, or plane is allowed to touch both pools.

Rev. 4 — Insufficient creepage distance under the optocoupler

Revision 3 also placed copper and silkscreen too close under and around the TLP250 package, well inside the creepage/clearance distance the datasheet specifies between the input and output leads for reliable isolation. Traces and copper pour on both sides of the barrier were pulled back to respect the datasheet's minimum spacing, keeping the isolation real in practice and not just on the schematic.

Rev. 5 — Gate-drive loop routed with too much area

In the same revision, the gate resistor and the IRF3205 gate/source pins were routed with a wide, looping trace instead of a short direct path, adding loop inductance right where the driver needs a clean, fast edge. The final layout tightens the TLP250-output-to-gate path and keeps the gate return next to it, following the low-inductance gate-loop layout advice in the TLP250 datasheet.

4. Result

The final board is a small, single-function isolation stage: PWM in on the control side, a clean gate drive out on the power side, and two ground pools that never touch except through light. It's a small module, but it's the one place in a power-switching stage where a shortcut on the ground layout is invisible until the MCU starts resetting itself every time the motor switches — which is exactly why it took five revisions to get right.